GRADE 7 ACCELERATED · UNIT 5

Unit 5 Review

Multi-Step Proportional Relationships

Name_____
Date_____
Period_____

Show your work. Circle your final answer.

HOW TO USE THIS. This review covers every learning target from the unit — more than the assessment does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

PART A. Scale Drawings

Lesson 1 · 5 questions

1. A 5 cm side of a figure is 8 cm in its scale copy. Find the scale factor.

2. A 16 cm side of a figure is 4 cm in its scale copy. Find the scale factor.

3. A scale copy is made at $k = 1.5$. A side of the original is 6 cm. Find the matching side of the copy.

4. A drawing has a 65° angle. Its scale copy is made at $k = 2$. What is the matching angle in the copy?

- ○ A. 32.5°
- ○ B. 67°
- ○ C. 130°
- ○ D. 65°

5. A map is drawn at $1\,\text{cm} : 30\,\text{km}$. Two towns are 4.5 cm apart on it. How far apart are they?

PART B. Scale Factor: Perimeter and Area

Lessons 2–3 · 6 questions

6. A figure with perimeter 16 cm is scaled by $k = 2.5$. Find the copy's perimeter.

7. A figure with area $9\,\text{cm}^2$ is scaled by $k = 2.5$. Find the copy's area.

8. A 4 cm by 6 cm rectangle is scaled by $k = \frac{1}{2}$.

a. Find the copy's perimeter.

b. Find the copy's area.

9. A figure has area $10\,\text{cm}^2$ and its scale copy has area $160\,\text{cm}^2$. Find k.

10. A figure's perimeter is 12 cm and its copy's perimeter is 36 cm. What is the area ratio, copy to original?

11. Sam says tripling every side of a rectangle triples its area. Explain why not, with a counterexample.

PART C. Real-World Scale Drawings

Lesson 4 · 4 questions

12. A plan is drawn at $\frac{1}{2}\,\text{in} : 1\,\text{ft}$. A wall is 7.5 in long on it. Find the real length.

13. A plan is drawn at $1\,\text{in} : 6\,\text{ft}$. A real wall is 21 ft long. Find its length on the plan.

14. A model boat is built at $1 : 25$ and is 32 cm long. Find the real boat's length in meters.

15. Every part below is about one floor plan drawn at $\frac{1}{4}\,\text{in} : 1\,\text{ft}$, on which a room is 3 in by 4 in. $1\,\text{ft} = 0.305\,\text{m}$.

a. Find the real length of the wall drawn 3 in long.

b. Find the room's real area.

c. Write the scale factor from the drawing to the room as a single number.

d. Find the room's real perimeter in meters.

PART D. Converting Length Across Systems; Currency

Lessons 6–7 · 6 questions

16. $1\,\text{in} = 2.54\,\text{cm}$. How many centimeters is 14 in?

17. $1\,\text{cm} = 0.39\,\text{in}$. How many inches is 120 cm?

18. $1\,\text{km} = 0.62\,\text{mi}$. How many miles is 30 km?

19. $1\,\text{yd} = 3\,\text{ft}$, $1\,\text{ft} = 12\,\text{in}$, $1\,\text{in} = 2.54\,\text{cm}$. How many centimeters is 6 yd? Estimate first.

20. Rate on Nov 3: $1\,\text{US dollar} = 0.92\,\text{euros}$.

a. How many euros is $150?

b. How many US dollars is 69 euros?

21. $1\,\text{in} = 2.54\,\text{cm}$; $1\,\text{cm} = 0.39\,\text{in}$. Lee converts 40 cm to inches by multiplying by 2.54 and gets 101.6 in. Name the error and fix it.

PART E. Converting Mass, Area and Volume

Lessons 8–9 · 5 questions

22. $1\,\text{lb} = 0.454\,\text{kg}$. How many kilograms is 80 lb?

23. $1\,\text{L} = 0.264\,\text{gal}$. How many gallons is 40 L?

24. $1\text{ ft} = 0.305\text{ m}$. How many square meters is 8 ft^2 ? (Round to two places.)

25. $1\text{ c} = 8\text{ fl oz}$ and $1\text{ c} = 0.24\text{ L}$. How many liters is 48 fl oz?

26. $1\text{ m} = 3.28\text{ ft}$. A desk is 1.5 m^2 in area. Which is closest to its area in square feet?

- ○ A. 4.9 ft^2
- ○ B. 9.8 ft^2
- ○ C. 16.1 ft^2
- ○ D. 10.8 ft^2

PART F. Multi-Step Proportion Problems

Lessons 10–11 · 4 questions

27. A car gets 29 miles per gallon and its tank holds 12 gal.

a. How far does it go on a full tank?

b. It starts an 800-mi trip with a full tank. How many tanks of gas does the trip take?

28. A recipe for 3 servings uses $1\frac{1}{4}$ cups of rice.

a. How much rice for 12 servings?

b. With 10 cups on hand, how many servings can be made?

29. $1\,\text{km} = 0.62\,\text{mi}$. A train does 110 km/h; a bus does 65 mi/h. Which is faster? Show both speeds in one unit.

30. Paint is mixed $1:2:5$. A batch uses 10 gal of the third color. How many gallons is the whole batch?

GRADE 7 ACCELERATED · UNIT 5

Unit 5 Review

Multi-Step Proportional Relationships **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

HOW TO USE THIS. This review covers every learning target from the unit — more than the assessment does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

PART A. Scale Drawings

Lesson 1 · MA.7.GR.1.5 · 5 questions

1. A 5 cm side of a figure is 8 cm in its scale copy. Find the scale factor.

k = 1.6

copy ÷ original

2. A 16 cm side of a figure is 4 cm in its scale copy. Find the scale factor.

k = 0.25

the copy is smaller, so k is under 1

3. A scale copy is made at $k = 1.5$. A side of the original is 6 cm. Find the matching side of the copy.

9 cm

4. A drawing has a 65° angle. Its scale copy is made at $k = 2$. What is the matching angle in the copy?

- ○ A. 32.5°
- ○ B. 67°
- ○ C. 130°
- ● **D. 65°**

D

5. A map is drawn at $1\,\text{cm} : 30\,\text{km}$. Two towns are 4.5 cm apart on it. How far apart are they?

135 km

drawing × rate

PART B. Scale Factor: Perimeter and Area

Lessons 2–3 · MA.7.GR.1.5 · 6 questions

6. A figure with perimeter 16 cm is scaled by $k = 2.5$. Find the copy's perimeter.

40 cm

perimeter × k

7. A figure with area $9\,\text{cm}^2$ is scaled by $k = 2.5$. Find the copy's area.

56.25 cm²

area × k^2 = 9 × 6.25

8. A 4 cm by 6 cm rectangle is scaled by $k = \frac{1}{2}$.

a. Find the copy's perimeter.

10 cm

20 × ½

b. Find the copy's area.

6 cm²

24 × ¼

9. A figure has area $10\,\text{cm}^2$ and its scale copy has area $160\,\text{cm}^2$. Find k.

k = 4

160 ÷ 10 = 16 = k^2

10. A figure's perimeter is 12 cm and its copy's perimeter is 36 cm. What is the area ratio, copy to original?

9

k = 3 from the perimeters; the area ratio is k^2

11. Sam says tripling every side of a rectangle triples its area. Explain why not, with a counterexample.

A 2 by 5 (area 10) tripled is 6 by 15 (area 90): nine times, because area is two lengths and each is tripled.

the reason and the numbers

PART C. Real-World Scale Drawings

Lesson 4 · MA.7.GR.1.5 · 4 questions

12. A plan is drawn at $\frac{1}{2}\,\text{in} : 1\,\text{ft}$. A wall is 7.5 in long on it. Find the real length.

15 ft

1 in : 2 ft; 7.5 × 2

13. A plan is drawn at $1\,\text{in} : 6\,\text{ft}$. A real wall is 21 ft long. Find its length on the plan.

3.5 in

real ÷ rate

14. A model boat is built at $1 : 25$ and is 32 cm long. Find the real boat's length in meters.

8 m

32 × 25 = 800 cm

15. SSDD. Every part below is about one floor plan drawn at ¼ in : 1 ft, on which a room is 3 in by 4 in. 1 ft = 0.305 m. Same surface, four different questions: a length, an area with the rate used twice, the scale factor as one number, and a length conversion — the four moves of the unit on one room, so that a student learns to read the question and not the picture.

a. Find the real length of the wall drawn 3 in long.

12 ft

1 in : 4 ft; 3 × 4

b. Find the room's real area.

192 ft²

12 by 16; the rate used twice

c. Write the scale factor from the drawing to the room as a single number.

48

1 in : 4 ft is 1 in : 48 in

d. Find the room's real perimeter in meters.

17.08 m

2(12 + 16) = 56 ft; 56 × 0.305 (56 ÷ 3.28 = 17.07 by the other ratio). The area in square meters is the test's transfer item and is not rehearsed here (ruling 18).

PART D. Converting Length Across Systems; Currency *Lessons 6–7 · MA.7.AR.3.3 · 6 questions*

16. $1\text{ in} = 2.54\text{ cm}$. How many centimeters is 14 in?

35.56 cm

17. $1\text{ cm} = 0.39\text{ in}$. How many inches is 120 cm?

46.8 in (47.2 by ÷ 2.54)

fewer inches than centimeters

18. $1\text{ km} = 0.62\text{ mi}$. How many miles is 30 km?

18.6 mi

19. $1\text{ yd} = 3\text{ ft}$, $1\text{ ft} = 12\text{ in}$, $1\text{ in} = 2.54\text{ cm}$. How many centimeters is 6 yd? Estimate first.

About 540; exactly 548.64 cm

216 in × 2.54

20. Rate on Nov 3: $1\text{ US dollar} = 0.92\text{ euros}$.

a. How many euros is $150?

138 euros

× the rate

b. How many US dollars is 69 euros?

$75

÷ the rate

21. $1\text{ in} = 2.54\text{ cm}$; $1\text{ cm} = 0.39\text{ in}$. Lee converts 40 cm to inches by multiplying by 2.54 and gets 101.6 in. Name the error and fix it.

Lee used the ratio whose 1 is inches on centimeters. Inches are bigger, so there are fewer: 40 × 0.39 = 15.6 in (or 40 ÷ 2.54 = 15.7).

the error named and the estimate

PART E. Converting Mass, Area and Volume

Lessons 8–9 · MA.7.AR.3.3 · 5 questions

22. $1\text{ lb} = 0.454\text{ kg}$. How many kilograms is 80 lb?

36.32 kg

(80 ÷ 2.204 = 36.30 by the other ratio)

23. $1\text{ L} = 0.264\text{ gal}$. How many gallons is 40 L?

10.56 gal

24. $1\text{ ft} = 0.305\text{ m}$. How many square meters is 8 ft^2 ? (Round to two places.)

About 0.74 m^2

$8 \times 0.305^2 = 0.7442$: the sheet's length ratio, squared

25. $1\text{ c} = 8\text{ fl oz}$ and $1\text{ c} = 0.24\text{ L}$. How many liters is 48 fl oz?

1.44 L

6 c × 0.24

26. $1\text{ m} = 3.28\text{ ft}$. A desk is 1.5 m^2 in area. Which is closest to its area in square feet?

- ○ A. 4.9 ft^2
- ○ B. 9.8 ft^2
- ● **C.** 16.1 ft^2
- ○ D. 10.8 ft^2

C

$1.5 \times 3.28^2 = 16.1376$.

PART F. Multi-Step Proportion Problems

Lessons 10–11 · MA.7.AR.4.5 · 4 questions

27. A car gets 29 miles per gallon and its tank holds 12 gal.

a. How far does it go on a full tank?

348 mi

b. It starts an 800-mi trip with a full tank. How many tanks of gas does the trip take?

3 tanks (800 ÷ 348 ≈ 2.3)

a part of a tank is still a tank

28. A recipe for 3 servings uses $1\frac{1}{4}$ cups of rice.

a. How much rice for 12 servings?

5 cups

k = 4

b. With 10 cups on hand, how many servings can be made?

24 servings

8 recipes × 3

29. $1\,\text{km} = 0.62\,\text{mi}$. A train does 110 km/h; a bus does 65 mi/h. Which is faster? Show both speeds in one unit.

The train: 110 km/h = 68.2 mi/h > 65 mi/h

same unit first

30. Paint is mixed $1:2:5$. A batch uses 10 gal of the third color. How many gallons is the whole batch?

16 gal

k = 2; 8 parts × 2

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form A · 24 points · two class periods

Name_____
Date_____
Period_____

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 10 cm in its scale copy. Find the scale factor. *1 pt*

2. A 15 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

3. A figure is scaled by a scale factor of 3. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. Each side is 3 times as long.
- ☐ B. Each angle is 3 times as large.
- ☐ C. The perimeter is 3 times as long.
- ☐ D. The area is 3 times as large.
- ☐ E. The area is 9 times as large.

4. A 3 cm by 4 cm rectangle is scaled by k = 3.

a. Find the copy's perimeter. *1 pt*

b. Find the copy's area. *1 pt*

5. A figure has area 8 cm^2. Its scale copy has area 72 cm^2. What is the scale factor? *1 pt*

6. A floor plan is drawn at ¼ in : 1 ft. A room is 3 in by 3.5 in on the plan.

a. Find the real length of the wall drawn 3 in long. *1 pt*

b. Find the room's real area. *1 pt*

7. A plan is drawn at 1 in : 4 ft. A real wall is 18 ft long. How long is it on the plan? *1 pt*

8. A plan is drawn at 1 in : 5 ft. A room is 2 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square meters. *1 pt*

Part 2. Converting Across Systems *8 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimeters is 9 in? *1 pt*

10. 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 70 cm. *1 pt*

- ☐ A. 70 × 0.39
- ☐ B. 70 × 2.54
- ☐ C. 70 ÷ 2.54
- ☐ D. 70 ÷ 0.39

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 4 yd? *1 pt*

12. 1 kg = 2.204 lb. How many pounds is 20 kg? *1 pt*

13. 1 gal = 3.785 L. How many liters is 5 gal? *1 pt*

14. 1 ft = 0.305 m. A patio is 20 ft^2 in area. Which is closest to its area in square meters? *1 pt*

- ○ A. 6.1 m^2
- ○ B. 0.093 m^2
- ○ C. 1.86 m^2
- ○ D. 12.2 m^2

15. Rate on Nov 5: 1 US dollar = 0.92 euros.

a. How many euros is $60? *1 pt*

b. How many US dollars is 23 euros? *1 pt*

Part 3. Multi-Step Proportion Problems *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car gets 26 miles per gallon and its tank holds 15 gal.

a. How far does it go on a full tank? *1 pt*

b. It makes a 1,000-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

17. A recipe for 4 servings uses $1\frac{1}{2}$ cups of rice.

a. How much rice for 10 servings? *1 pt*

b. With 6 cups on hand, how many servings can be made? *1 pt*

18. Juice is mixed orange : mango : water = 3 : 2 : 4. A batch uses 9 cups of orange. How many cups is the whole batch? *1 pt*

19. 1 mi = 1.61 km. A drive is 200 km long. Car A averages 40 mi/h; car B averages 60 km/h. Which car gets there first, and by about how many minutes? Show both speeds in one unit. *1 pt*

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form A · 24 points · two class periods **ANSWER KEY**

Name
Date
Period

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points MA.7.GR.1.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 10 cm in its scale copy. Find the scale factor. *1 pt*

k = 2.5

copy ÷ original

Scoring: 1 point.

2. A 15 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.4

copy ÷ original; the copy is smaller, so k is under 1 — 2.5 is the proportion backwards

Scoring: 1 point.

3. A figure is scaled by a scale factor of 3. Select ALL of the statements that are true about the copy. *1 pt*

- ☒ **A. Each side is 3 times as long.**
- ☐ B. Each angle is 3 times as large.
- ☒ **C. The perimeter is 3 times as long.**
- ☐ D. The area is 3 times as large.
- ☒ **E. The area is 9 times as large.**

A, C and E

All three and nothing else for the point: sides and perimeter × k, area × k^2, angles unchanged.

Scoring: 1 point.

4. A 3 cm by 4 cm rectangle is scaled by k = 3.

a. Find the copy's perimeter. *1 pt*

42 cm

perimeter × k

b. Find the copy's area. *1 pt*

108 cm²

area × k^2 — 36 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 8 cm². Its scale copy has area 72 cm². What is the scale factor? *1 pt*

k = 3

72 ÷ 8 = 9 is k^2, so k = √9. A board with 9 handed in the area ratio as k.

Scoring: 1 point.

6. A floor plan is drawn at ¼ in : 1 ft. A room is 3 in by 3.5 in on the plan.

a. Find the real length of the wall drawn 3 in long. *1 pt*

12 ft

the scale per 1 inch, then ×: 1 in : 4 ft

b. Find the room's real area. *1 pt*

168 ft²

real dimensions first: 12 by 14 — 10.5 in^2 × 4 = 42 is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at 1 in : 4 ft. A real wall is 18 ft long. How long is it on the plan? *1 pt*

4.5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 5 ft. A room is 2 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square meters. *1 pt*

About 9.30 m² (3.05 m by 3.05 m); 9.29 by the sheet's other ratio

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. 10 ft by 10 ft; 3.05 × 3.05 = 9.3025. The same number by the other route: 100 ft^2 × 0.305^2 = 9.3025; by the sheet's 1 m = 3.28 ft, 100 ÷ 3.28^2 = 9.29 — full credit. 30.5 (100 × 0.305) used the ratio once on an area. A plan's area asked in metric appears on no review and in no question bank; its type was 5.10 board 8.

Scoring: 1 point.

Part 2. Converting Across Systems

8 points *MA.7.AR.3.3*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimeters is 9 in? *1 pt*

22.86 cm (23.1 by ÷ 0.39)

× the ratio whose 1 is inches; the sheet's other ratio, ÷ 0.39, earns the point too when the work shows it

Scoring: 1 point.

10. 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 70 cm. *1 pt*

- ☒ **A. 70 × 0.39**
- ☐ B. 70 × 2.54
- ☒ **C. 70 ÷ 2.54**
- ☐ D. 70 ÷ 0.39

A and C

Both sheet routes (27.3 and 27.6) and nothing else for the point.

Scoring: 1 point.

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 4 yd? *1 pt*

365.76 cm

a chain: 144 in, then × 2.54 (144 ÷ 0.39 = 369.2 by the other ratio — full credit). The estimate (a yard is nearly a meter) catches 10.16.

Scoring: 1 point.

12. 1 kg = 2.204 lb. How many pounds is 20 kg? *1 pt*

44.08 lb (44.05 by ÷ 0.454)

× the ratio whose 1 is the unit you have; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

13. 1 gal = 3.785 L. How many liters is 5 gal? *1 pt*

18.925 L (18.94 by ÷ 0.264)

more liters than gallons; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

14. 1 ft = 0.305 m. A patio is 20 ft^2 in area. Which is closest to its area in square meters? *1 pt*

- ○ A. 6.1 m^2
- ○ B. 0.093 m^2
- ● **C. 1.86 m^2**
- ○ D. 12.2 m^2

C

20 × 0.305^2 = 1.8605 (or 20 ÷ 3.28^2 = 1.86).

Scoring: 1 point.

15. Rate on Nov 5: 1 US dollar = 0.92 euros.

a. How many euros is $60? *1 pt*

55.20 euros

× the rate

b. How many US dollars is 23 euros? *1 pt*

$25

÷ the rate; 21.16 multiplied both ways

Scoring: a 1 · b 1 = 2 points.

Part 3. Multi-Step Proportion Problems

6 points MA.7.AR.4.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car gets 26 miles per gallon and its tank holds 15 gal.

a. How far does it go on a full tank? *1 pt*

390 mi

the rate × the gallons

b. It makes a 1,000-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

3 tanks (1,000 ÷ 390 ≈ 2.56)

the bracket on the paper settles the rounding; follow-through on part (a)

Scoring: a 1 · b 1 = 2 points.

17. A recipe for 4 servings uses $1\frac{1}{2}$ cups of rice.

a. How much rice for 10 servings? *1 pt*

3¾ cups

k = 2.5 from the yields, then the ingredient

b. With 6 cups on hand, how many servings can be made? *1 pt*

16 servings

4 recipes × the yield; 24 scaled the yield by the cups

Scoring: a 1 · b 1 = 2 points.

18. Juice is mixed orange : mango : water = 3 : 2 : 4. A batch uses 9 cups of orange. How many cups is the whole batch? *1 pt*

27 cups

k = 3 from the orange, then the whole, 9 parts; 6 found the mango and stopped

Scoring: 1 point.

19. 1 mi = 1.61 km. A drive is 200 km long. Car A averages 40 mi/h; car B averages 60 km/h. Which car gets there first, and by about how many minutes? Show both speeds in one unit. *1 pt*

Car A, by about 14 minutes: 40 mi/h = 64.4 km/h, so 200 ÷ 64.4 ≈ 3.11 h against 200 ÷ 60 ≈ 3.33 h.

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. The point needs the speeds in one unit and the choice with its gap: 0.227 h is 13.7 min — 13 or 14 both earn it (by the sheet's 0.62 route, 124 mi: 3.1 h against 3.33 h, 14 min). 'Car B, 60 > 40' compared two units. A decision from two rates in different units appears on no review and in no question bank; its type was 5.10 board 9.

Scoring: 1 point.

Score Tracker

Benchmark	Questions	Possible	Earned
MA.7.GR.1.5	1, 2, 3, 4, 5, 6, 7, 8	10	
MA.7.AR.3.3	9, 10, 11, 12, 13, 14, 15	8	
MA.7.AR.4.5	16, 17, 18, 19	6	
Total		**24**	

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form A · 24 points · two class periods **WORKED ANSWERS**

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors *10 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 10 cm in its scale copy. Find the scale factor. *1 pt*

k = 2.5

copy ÷ original

Scoring: 1 point.

2. A 15 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.4

copy ÷ original; the copy is smaller, so k is under 1 — 2.5 is the proportion backwards

Scoring: 1 point.

3. A figure is scaled by a scale factor of 3. Select ALL of the statements that are true about the copy. *1 pt*

- ☒ **A. Each side is 3 times as long.**
- ☐ B. Each angle is 3 times as large.
- ☒ **C. The perimeter is 3 times as long.**
- ☐ D. The area is 3 times as large.
- ☒ **E. The area is 9 times as large.**

A, C and E

All three and nothing else for the point: sides and perimeter × k, area × k^2, angles unchanged.

Scoring: 1 point.

4. A 3 cm by 4 cm rectangle is scaled by k = 3.

a. Find the copy's perimeter. *1 pt*

42 cm

perimeter × k

b. Find the copy's area. *1 pt*

108 cm²

area × k² — 36 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 8 cm². Its scale copy has area 72 cm². What is the scale factor? *1 pt*

k = 3

72 ÷ 8 = 9 is k², so k = √9. A board with 9 handed in the area ratio as k.

Scoring: 1 point.

6. A floor plan is drawn at ¼ in : 1 ft. A room is 3 in by 3.5 in on the plan.

a. Find the real length of the wall drawn 3 in long. *1 pt*

12 ft

the scale per 1 inch, then ×: 1 in : 4 ft

b. Find the room's real area. *1 pt*

168 ft²

real dimensions first: 12 by 14 — 10.5 in² × 4 = 42 is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at 1 in : 4 ft. A real wall is 18 ft long. How long is it on the plan? *1 pt*

4.5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 5 ft. A room is 2 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square meters. *1 pt*

About 9.30 m² (3.05 m by 3.05 m); 9.29 by the sheet's other ratio

10 ft by 10 ft; 3.05 × 3.05 = 9.3025. The same number by the other route: 100 ft² × 0.305² = 9.3025; by the sheet's 1 m = 3.28 ft, 100 ÷ 3.28² = 9.29 — full credit. 30.5 (100 × 0.305) used the ratio once on an area. A plan's area asked in metric appears on no review and in no question bank; its type was 5.10 board 8.

Scoring: 1 point.

Part 2. Converting Across Systems *8 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimeters is 9 in? *1 pt*

22.86 cm (23.1 by ÷ 0.39)

× the ratio whose 1 is inches; the sheet's other ratio, ÷ 0.39, earns the point too when the work shows it

Scoring: 1 point.

10. 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 70 cm. *1 pt*

- ☒ **A. 70 × 0.39**
- ☐ B. 70 × 2.54
- ☒ **C. 70 ÷ 2.54**
- ☐ D. 70 ÷ 0.39

A and C

Both sheet routes (27.3 and 27.6) and nothing else for the point.

Scoring: 1 point.

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 4 yd? *1 pt*

365.76 cm

a chain: 144 in, then × 2.54 (144 ÷ 0.39 = 369.2 by the other ratio — full credit). The estimate (a yard is nearly a meter) catches 10.16.

Scoring: 1 point.

12. 1 kg = 2.204 lb. How many pounds is 20 kg? *1 pt*

44.08 lb (44.05 by ÷ 0.454)

× the ratio whose 1 is the unit you have; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

13. 1 gal = 3.785 L. How many liters is 5 gal? *1 pt*

18.925 L (18.94 by ÷ 0.264)

more liters than gallons; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

14. 1 ft = 0.305 m. A patio is 20 ft^2 in area. Which is closest to its area in square meters? *1 pt*

- ○ A. 6.1 m^2
- ○ B. 0.093 m^2
- ● **C. 1.86 m^2**
- ○ D. 12.2 m^2

C

20 × 0.305^2 = 1.8605 (or 20 ÷ 3.28^2 = 1.86).

Scoring: 1 point.

15. Rate on Nov 5: 1 US dollar = 0.92 euros.

a. How many euros is $60? *1 pt*

55.20 euros

× the rate

b. How many US dollars is 23 euros? *1 pt*

$25

÷ the rate; 21.16 multiplied both ways

Scoring: a 1 · b 1 = 2 points.

Part 3. Multi-Step Proportion Problems *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car gets 26 miles per gallon and its tank holds 15 gal.

a. How far does it go on a full tank? *1 pt*

390 mi

the rate × the gallons

b. It makes a 1,000-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

3 tanks (1,000 ÷ 390 ≈ 2.56)

the bracket on the paper settles the rounding; follow-through on part (a)

Scoring: a 1 · b 1 = 2 points.

17. A recipe for 4 servings uses $1\frac{1}{2}$ cups of rice.

a. How much rice for 10 servings? *1 pt*

3¾ cups

k = 2.5 from the yields, then the ingredient

b. With 6 cups on hand, how many servings can be made? *1 pt*

16 servings

4 recipes × the yield; 24 scaled the yield by the cups

Scoring: a 1 · b 1 = 2 points.

18. Juice is mixed orange : mango : water = 3 : 2 : 4. A batch uses 9 cups of orange. How many cups is the whole batch? *1 pt*

27 cups

k = 3 from the orange, then the whole, 9 parts; 6 found the mango and stopped

Scoring: 1 point.

19. 1 mi = 1.61 km. A drive is 200 km long. Car A averages 40 mi/h; car B averages 60 km/h. Which car gets there first, and by about how many minutes? Show both speeds in one unit. *1 pt*

Car A, by about 14 minutes: 40 mi/h = 64.4 km/h, so 200 ÷ 64.4 ≈ 3.11 h against 200 ÷ 60 ≈ 3.33 h.

The point needs the speeds in one unit and the choice with its gap: 0.227 h is 13.7 min — 13 or 14 both earn it (by the sheet's 0.62 route, 124 mi: 3.1 h against 3.33 h, 14 min). 'Car B, 60 > 40' compared two units. A decision from two rates in different units appears on no review and in no question bank; its type was 5.10 board 9.

Scoring: 1 point.

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form B · 24 points · two class periods

Name_____
Date_____
Period_____

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 6 cm side of a figure is 9 cm in its scale copy. Find the scale factor. *1 pt*

2. A 20 cm side of a figure is 5 cm in its scale copy. Find the scale factor. *1 pt*

3. A figure is scaled by a scale factor of 4. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. Each angle is 4 times as large.
- ☐ B. The area is 16 times as large.
- ☐ C. Each side is 4 times as long.
- ☐ D. The area is 4 times as large.
- ☐ E. The perimeter is 4 times as long.

4. A 2 cm by 5 cm rectangle is scaled by k = 4.

a. Find the copy's perimeter. *1 pt*

b. Find the copy's area. *1 pt*

5. A figure has area 5 cm^2. Its scale copy has area 80 cm^2. What is the scale factor? *1 pt*

6. A floor plan is drawn at ½ in : 1 ft. A room is 3 in by 5 in on the plan.

a. Find the real length of the wall drawn 3 in long. *1 pt*

b. Find the room's real area. *1 pt*

7. A plan is drawn at ¼ in : 1 ft. A real wall is 22 ft long. How long is it on the plan? *1 pt*

8. A plan is drawn at 1 in : 4 ft. A room is 1.5 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square meters. *1 pt*

Part 2. Converting Across Systems *8 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimeters is 11 in? *1 pt*

10. 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 90 cm. *1 pt*

- ☐ A. 90 ÷ 0.39
- ☐ B. 90 ÷ 2.54
- ☐ C. 90 × 2.54
- ☐ D. 90 × 0.39

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 5 yd? *1 pt*

12. 1 lb = 0.454 kg. How many kilograms is 60 lb? *1 pt*

13. 1 gal = 3.785 L. How many liters is 9 gal? *1 pt*

14. 1 m = 3.28 ft. A floor is 5 m^2 in area. Which is closest to its area in square feet? *1 pt*

- ○ A. 53.8 ft^2
- ○ B. 16.4 ft^2
- ○ C. 10.8 ft^2
- ○ D. 32.8 ft^2

15. Rate on Nov 5: 1 US dollar = 1.35 Canadian dollars.

a. How many Canadian dollars is $70? *1 pt*

b. How many US dollars is 54 Canadian dollars? *1 pt*

Part 3. Multi-Step Proportion Problems *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car gets 22 miles per gallon and its tank holds 20 gal.

a. How far does it go on a full tank? *1 pt*

b. It makes a 1,500-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

17. A recipe for 6 servings uses 2 cups of rice.

a. How much rice for 9 servings? *1 pt*

b. With 8 cups on hand, how many servings can be made? *1 pt*

18. Paint is mixed 1 : 3 : 4. A batch uses 12 gal of the second color. How many gallons is the whole batch? *1 pt*

19. 1 mi = 1.61 km. A drive is 450 km long. Truck A averages 55 mi/h; truck B averages 90 km/h. Which truck gets there first, and by about how many minutes? Show both speeds in one unit. *1 pt*

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form B · 24 points · two class periods **ANSWER KEY**

Name
Date
Period

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points MA.7.GR.1.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 6 cm side of a figure is 9 cm in its scale copy. Find the scale factor. *1 pt*

k = 1.5

copy ÷ original

Scoring: 1 point.

2. A 20 cm side of a figure is 5 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.25

copy ÷ original; the copy is smaller, so k is under 1 — 4 is the proportion backwards

Scoring: 1 point.

3. A figure is scaled by a scale factor of 4. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. Each angle is 4 times as large.
- ☒ **B. The area is 16 times as large.**
- ☒ **C. Each side is 4 times as long.**
- ☐ D. The area is 4 times as large.
- ☒ **E. The perimeter is 4 times as long.**

B, C and E

All three and nothing else for the point: sides and perimeter × k, area × k^2, angles unchanged.

Scoring: 1 point.

4. A 2 cm by 5 cm rectangle is scaled by k = 4.

a. Find the copy's perimeter. *1 pt*

56 cm

perimeter × k

b. Find the copy's area. *1 pt*

160 cm²

area × k² — 40 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 5 cm². Its scale copy has area 80 cm². What is the scale factor? *1 pt*

k = 4

80 ÷ 5 = 16 is k², so k = √16. A board with 16 handed in the area ratio as k.

Scoring: 1 point.

6. A floor plan is drawn at ½ in : 1 ft. A room is 3 in by 5 in on the plan.

a. Find the real length of the wall drawn 3 in long. *1 pt*

6 ft

the scale per 1 inch, then ×: 1 in : 2 ft

b. Find the room's real area. *1 pt*

60 ft²

real dimensions first: 6 by 10 — 15 in² × 2 = 30 is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at ¼ in : 1 ft. A real wall is 22 ft long. How long is it on the plan? *1 pt*

5.5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 4 ft. A room is 1.5 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square meters. *1 pt*

About 4.47 m² (1.83 m by 2.44 m); 4.46 by the sheet's other ratio

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. 6 ft by 8 ft; 1.83 × 2.44 = 4.4652. The same number by the other route: 48 ft² × 0.305² = 4.4652; by the sheet's 1 m = 3.28 ft, 48 ÷ 3.28² = 4.46 — full credit. 14.64 (48 × 0.305) used the ratio once on an area. A plan's area asked in metric appears on no review and in no question bank; its type was 5.10 board 8.

Scoring: 1 point.

Part 2. Converting Across Systems

8 points *MA.7.AR.3.3*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimeters is 11 in? *1 pt*

27.94 cm (28.2 by ÷ 0.39)

× the ratio whose 1 is inches; the sheet's other ratio, ÷ 0.39, earns the point too when the work shows it

Scoring: 1 point.

10. 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 90 cm. *1 pt*

- ☐ A. 90 ÷ 0.39
- ☒ **B. 90 ÷ 2.54**
- ☐ C. 90 × 2.54
- ☒ **D. 90 × 0.39**

B and D

Both sheet routes (35.1 and 35.4) and nothing else for the point.

Scoring: 1 point.

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 5 yd? *1 pt*

457.2 cm

a chain: 180 in, then × 2.54 (180 ÷ 0.39 = 461.5 by the other ratio — full credit). The estimate catches 12.7.

Scoring: 1 point.

12. 1 lb = 0.454 kg. How many kilograms is 60 lb? *1 pt*

27.24 kg (27.22 by ÷ 2.204)

× the ratio whose 1 is the unit you have; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

13. 1 gal = 3.785 L. How many liters is 9 gal? *1 pt*

34.065 L (34.09 by ÷ 0.264)

more liters than gallons; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

14. 1 m = 3.28 ft. A floor is 5 m^2 in area. Which is closest to its area in square feet? *1 pt*

- ● **A. 53.8 ft^2**
- ○ B. 16.4 ft^2
- ○ C. 10.8 ft^2
- ○ D. 32.8 ft^2

A

5 × 3.28^2 = 53.79 (or 5 ÷ 0.305^2 = 53.75).

Scoring: 1 point.

15. Rate on Nov 5: 1 US dollar = 1.35 Canadian dollars.

a. How many Canadian dollars is $70? *1 pt*

94.50 Canadian dollars

× the rate

b. How many US dollars is 54 Canadian dollars? *1 pt*

$40

÷ the rate; 72.90 multiplied both ways

Scoring: a 1 · b 1 = 2 points.

Part 3. Multi-Step Proportion Problems

6 points MA.7.AR.4.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car gets 22 miles per gallon and its tank holds 20 gal.

a. How far does it go on a full tank? *1 pt*

440 mi

the rate × the gallons

b. It makes a 1,500-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

4 tanks (1,500 ÷ 440 ≈ 3.41)

the bracket on the paper settles the rounding; follow-through on part (a)

Scoring: a 1 · b 1 = 2 points.

17. A recipe for 6 servings uses 2 cups of rice.

a. How much rice for 9 servings? *1 pt*

3 cups

k = 1.5 from the yields, then the ingredient

b. With 8 cups on hand, how many servings can be made? *1 pt*

24 servings

4 recipes × the yield; 48 scaled the yield by the cups

Scoring: a 1 · b 1 = 2 points.

18. Paint is mixed 1 : 3 : 4. A batch uses 12 gal of the second color. How many gallons is the whole batch? *1 pt*

32 gal

k = 4 from the second color, then the whole, 8 parts; 16 found the third color and stopped

Scoring: 1 point.

19. 1 mi = 1.61 km. A drive is 450 km long. Truck A averages 55 mi/h; truck B averages 90 km/h. Which truck gets there first, and by about how many minutes? Show both speeds in one unit. *1 pt*

Truck B, by about 5 minutes: 55 mi/h = 88.55 km/h, so 450 ÷ 90 = 5 h against 450 ÷ 88.55 ≈ 5.08 h.

TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. The point needs the speeds in one unit and the choice with its gap: 0.082 h is 4.9 min — 5 earns it (by the 0.62 route, 279 mi: 5 h against 5.07 h, 4 min; accept 4–5). 'Truck B, 90 > 55' is right by accident — no credit without the conversion. The surface appears on no review and in no question bank; its type was 5.10 board 9.

Scoring: 1 point.

Score Tracker

Benchmark	Questions	Possible	Earned
MA.7.GR.1.5	1, 2, 3, 4, 5, 6, 7, 8	10	
MA.7.AR.3.3	9, 10, 11, 12, 13, 14, 15	8	
MA.7.AR.4.5	16, 17, 18, 19	6	
Total		**24**	

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.

GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form B · 24 points · two class periods **WORKED ANSWERS**

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors *10 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 6 cm side of a figure is 9 cm in its scale copy. Find the scale factor. *1 pt*

k = 1.5

copy ÷ original

Scoring: 1 point.

2. A 20 cm side of a figure is 5 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.25

copy ÷ original; the copy is smaller, so k is under 1 — 4 is the proportion backwards

Scoring: 1 point.

3. A figure is scaled by a scale factor of 4. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. Each angle is 4 times as large.
- ☒ **B. The area is 16 times as large.**
- ☒ **C. Each side is 4 times as long.**
- ☐ D. The area is 4 times as large.
- ☒ **E. The perimeter is 4 times as long.**

B, C and E

All three and nothing else for the point: sides and perimeter × k, area × k^2, angles unchanged.

Scoring: 1 point.

4. A 2 cm by 5 cm rectangle is scaled by k = 4.

a. Find the copy's perimeter. *1 pt*

56 cm

perimeter × k

b. Find the copy's area. *1 pt*

160 cm²

area × k² — 40 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 5 cm². Its scale copy has area 80 cm². What is the scale factor? *1 pt*

k = 4

80 ÷ 5 = 16 is k², so k = √16. A board with 16 handed in the area ratio as k.

Scoring: 1 point.

6. A floor plan is drawn at ½ in : 1 ft. A room is 3 in by 5 in on the plan.

a. Find the real length of the wall drawn 3 in long. *1 pt*

6 ft

the scale per 1 inch, then ×: 1 in : 2 ft

b. Find the room's real area. *1 pt*

60 ft²

real dimensions first: 6 by 10 — 15 in² × 2 = 30 is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at ¼ in : 1 ft. A real wall is 22 ft long. How long is it on the plan? *1 pt*

5.5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 4 ft. A room is 1.5 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square meters. *1 pt*

About 4.47 m² (1.83 m by 2.44 m); 4.46 by the sheet's other ratio

6 ft by 8 ft; 1.83 × 2.44 = 4.4652. The same number by the other route: 48 ft² × 0.305² = 4.4652; by the sheet's 1 m = 3.28 ft, 48 ÷ 3.28² = 4.46 — full credit. 14.64 (48 × 0.305) used the ratio once on an area. A plan's area asked in metric appears on no review and in no question bank; its type was 5.10 board 8.

Scoring: 1 point.

Part 2. Converting Across Systems

8 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimeters is 11 in? *1 pt*

27.94 cm (28.2 by ÷ 0.39)

× the ratio whose 1 is inches; the sheet's other ratio, ÷ 0.39, earns the point too when the work shows it

Scoring: 1 point.

10. 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 90 cm. *1 pt*

- ☐ A. 90 ÷ 0.39
- ☒ **B. 90 ÷ 2.54**
- ☐ C. 90 × 2.54
- ☒ **D. 90 × 0.39**

B and D

Both sheet routes (35.1 and 35.4) and nothing else for the point.

Scoring: 1 point.

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 5 yd? *1 pt*

457.2 cm

a chain: 180 in, then × 2.54 (180 ÷ 0.39 = 461.5 by the other ratio — full credit). The estimate catches 12.7.

Scoring: 1 point.

12. 1 lb = 0.454 kg. How many kilograms is 60 lb? *1 pt*

27.24 kg (27.22 by ÷ 2.204)

× the ratio whose 1 is the unit you have; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

13. 1 gal = 3.785 L. How many liters is 9 gal? *1 pt*

34.065 L (34.09 by ÷ 0.264)

more liters than gallons; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

14. 1 m = 3.28 ft. A floor is 5 m^2 in area. Which is closest to its area in square feet? *1 pt*

- ● **A. 53.8 ft^2**
- ○ B. 16.4 ft^2
- ○ C. 10.8 ft^2
- ○ D. 32.8 ft^2

A

5 × 3.28^2 = 53.79 (or 5 ÷ 0.305^2 = 53.75).

Scoring: 1 point.

15. Rate on Nov 5: 1 US dollar = 1.35 Canadian dollars.

a. How many Canadian dollars is $70? *1 pt*

94.50 Canadian dollars

× the rate

b. How many US dollars is 54 Canadian dollars? *1 pt*

$40

÷ the rate; 72.90 multiplied both ways

Scoring: a 1 · b 1 = 2 points.

Part 3. Multi-Step Proportion Problems *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car gets 22 miles per gallon and its tank holds 20 gal.

a. How far does it go on a full tank? *1 pt*

440 mi

the rate × the gallons

b. It makes a 1,500-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

4 tanks (1,500 ÷ 440 ≈ 3.41)

the bracket on the paper settles the rounding; follow-through on part (a)

Scoring: a 1 · b 1 = 2 points.

17. A recipe for 6 servings uses 2 cups of rice.

a. How much rice for 9 servings? *1 pt*

3 cups

k = 1.5 from the yields, then the ingredient

b. With 8 cups on hand, how many servings can be made? *1 pt*

24 servings

4 recipes × the yield; 48 scaled the yield by the cups

Scoring: a 1 · b 1 = 2 points.

18. Paint is mixed 1 : 3 : 4. A batch uses 12 gal of the second color. How many gallons is the whole batch? *1 pt*

32 gal

k = 4 from the second color, then the whole, 8 parts; 16 found the third color and stopped

Scoring: 1 point.

19. 1 mi = 1.61 km. A drive is 450 km long. Truck A averages 55 mi/h; truck B averages 90 km/h. Which truck gets there first, and by about how many minutes? Show both speeds in one unit. *1 pt*

Truck B, by about 5 minutes: 55 mi/h = 88.55 km/h, so 450 ÷ 90 = 5 h against 450 ÷ 88.55 ≈ 5.08 h.

The point needs the speeds in one unit and the choice with its gap: 0.082 h is 4.9 min — 5 earns it (by the 0.62 route, 279 mi: 5 h against 5.07 h, 4 min; accept 4–5). 'Truck B, 90 > 55' is right by accident — no credit without the conversion. The surface appears on no review and in no question bank; its type was 5.10 board 9.

Scoring: 1 point.

GRADE 7 ACCELERATED · UNIT 5 · PRACTICE TEST

Multi-Step Proportional Relationships

24 points

Name_____
Date_____
Period_____

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 12 cm in its scale copy. Find the scale factor. *1 pt*

2. A 12 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

3. A figure is scaled by a scale factor of 2. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. The area is 2 times as large.
- ☐ B. Each side is 2 times as long.
- ☐ C. The perimeter is 2 times as long.
- ☐ D. The area is 4 times as large.
- ☐ E. Each angle is 2 times as large.

4. A 3 cm by 5 cm rectangle is scaled by k = 2.

a. Find the copy's perimeter. *1 pt*

b. Find the copy's area. *1 pt*

5. A figure has area 6 cm^2. Its scale copy has area 150 cm^2. What is the scale factor? *1 pt*

6. A floor plan is drawn at 1 in : 5 ft. A room is 2 in by 3 in on the plan.

a. Find the real length of the wall drawn 2 in long. *1 pt*

b. Find the room's real area. *1 pt*

7. A plan is drawn at 1 in : 3 ft. A real wall is 15 ft long. How long is it on the plan? *1 pt*

8. A plan is drawn at 1 in : 6 ft. A room is 1.5 in by 2.5 in on the plan. Find the room's real area. *1 pt*

Part 2. Converting Across Systems *8 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimeters is 7 in? *1 pt*

10. 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 110 cm. *1 pt*

- ☐ A. 110 × 2.54
- ☐ B. 110 × 0.39
- ☐ C. 110 ÷ 0.39
- ☐ D. 110 ÷ 2.54

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 1.5 yd? *1 pt*

12. 1 kg = 2.204 lb. How many pounds is 35 kg? *1 pt*

13. 1 gal = 3.785 L. How many liters is 3 gal? *1 pt*

14. 1 in = 2.54 cm. A tile is 10 in^2 in area. Which is closest to its area in square centimeters? *1 pt*

- ○ A. 25.4 cm^2
- ○ B. 64.5 cm^2
- ○ C. 6.45 cm^2
- ○ D. 50.8 cm^2

15. Rate on Nov 5: 1 US dollar = 1.32 Canadian dollars.

a. How many Canadian dollars is $90? *1 pt*

b. How many US dollars is 66 Canadian dollars? *1 pt*

Part 3. Multi-Step Proportion Problems *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car gets 27 miles per gallon and its tank holds 10 gal.

a. How far does it go on a full tank? *1 pt*

b. It makes a 1,200-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

17. A recipe for 5 servings uses 2 cups of rice.

a. How much rice for 15 servings? *1 pt*

b. With 10 cups on hand, how many servings can be made? *1 pt*

18. Paint is mixed 3 : 1 : 2. A batch uses 4 gal of the second color. How many gallons is the whole batch? *1 pt*

19. 1 mi = 1.61 km. A train does 80 km/h; a bus does 45 mi/h. Which is faster? Show both speeds in one unit. *1 pt*

GRADE 7 ACCELERATED · UNIT 5 · PRACTICE TEST

Multi-Step Proportional Relationships

24 points **ANSWER KEY**

Name
Date
Period

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points MA.7.GR.1.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 12 cm in its scale copy. Find the scale factor. *1 pt*

k = 3

copy ÷ original

Scoring: 1 point.

2. A 12 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.5

copy ÷ original; the copy is smaller, so k is under 1 — 2 is the proportion backwards

Scoring: 1 point.

3. A figure is scaled by a scale factor of 2. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. The area is 2 times as large.
- ☒ **B. Each side is 2 times as long.**
- ☒ **C. The perimeter is 2 times as long.**
- ☒ **D. The area is 4 times as large.**
- ☐ E. Each angle is 2 times as large.

B, C and D

All three and nothing else for the point: sides and perimeter × k, area × k^2, angles unchanged.

Scoring: 1 point.

4. A 3 cm by 5 cm rectangle is scaled by k = 2.

a. Find the copy's perimeter. *1 pt*

32 cm

perimeter × k

b. Find the copy's area. *1 pt*

60 cm²

area × k² — 30 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 6 cm². Its scale copy has area 150 cm². What is the scale factor? *1 pt*

k = 5

150 ÷ 6 = 25 is k², so k = √25. A board with 25 handed in the area ratio as k.

Scoring: 1 point.

6. A floor plan is drawn at 1 in : 5 ft. A room is 2 in by 3 in on the plan.

a. Find the real length of the wall drawn 2 in long. *1 pt*

10 ft

the scale per 1 inch, then ×: 1 in : 5 ft

b. Find the room's real area. *1 pt*

150 ft²

real dimensions first: 10 by 15 — 6 in² × 5 = 30 is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at 1 in : 3 ft. A real wall is 15 ft long. How long is it on the plan? *1 pt*

5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 6 ft. A room is 1.5 in by 2.5 in on the plan. Find the room's real area. *1 pt*

135 ft²

9 ft by 15 ft.

Scoring: 1 point.

Part 2. Converting Across Systems

8 points *MA.7.AR.3.3*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimeters is 7 in? *1 pt*

17.78 cm (17.9 by ÷ 0.39)

× the ratio whose 1 is inches; the sheet's other ratio, ÷ 0.39, earns the point too when the work shows it

Scoring: 1 point.

10. 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 110 cm. *1 pt*

- ☐ A. 110 × 2.54
- ☒ **B. 110 × 0.39**
- ☐ C. 110 ÷ 0.39
- ☒ **D. 110 ÷ 2.54**

B and D

Both sheet routes (42.9 and 43.3) and nothing else for the point.

Scoring: 1 point.

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 1.5 yd? *1 pt*

137.16 cm

a chain: 54 in, then × 2.54 (54 ÷ 0.39 = 138.5 by the other ratio — full credit). The estimate catches 3.81.

Scoring: 1 point.

12. 1 kg = 2.204 lb. How many pounds is 35 kg? *1 pt*

77.14 lb (77.09 by ÷ 0.454)

× the ratio whose 1 is the unit you have; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

13. 1 gal = 3.785 L. How many liters is 3 gal? *1 pt*

11.355 L (11.36 by ÷ 0.264)

more liters than gallons; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

14. 1 in = 2.54 cm. A tile is 10 in^2 in area. Which is closest to its area in square centimeters? *1 pt*

- ○ A. 25.4 cm^2
- ● **B. 64.5 cm^2**
- ○ C. 6.45 cm^2
- ○ D. 50.8 cm^2

B

$10 \times 2.54^2 = 64.52$ (or $10 \div 0.39^2 = 65.7$ — the sheet's two routes differ here; either picks B).

Scoring: 1 point.

15. Rate on Nov 5: 1 US dollar = 1.32 Canadian dollars.

a. How many Canadian dollars is $90? *1 pt*

118.80 Canadian dollars

× the rate

b. How many US dollars is 66 Canadian dollars? *1 pt*

$50

÷ the rate; 87.12 multiplied both ways

Scoring: a 1 · b 1 = 2 points.

Part 3. Multi-Step Proportion Problems

6 points MA.7.AR.4.5

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car gets 27 miles per gallon and its tank holds 10 gal.

a. How far does it go on a full tank? *1 pt*

270 mi

the rate × the gallons

b. It makes a 1,200-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

5 tanks (1,200 ÷ 270 ≈ 4.44)

the bracket on the paper settles the rounding; follow-through on part (a)

Scoring: a 1 · b 1 = 2 points.

17. A recipe for 5 servings uses 2 cups of rice.

a. How much rice for 15 servings? *1 pt*

6 cups

k = 3 from the yields, then the ingredient

b. With 10 cups on hand, how many servings can be made? *1 pt*

25 servings

5 recipes × the yield; 50 scaled the yield by the cups

Scoring: a 1 · b 1 = 2 points.

18. Paint is mixed 3 : 1 : 2. A batch uses 4 gal of the second color. How many gallons is the whole batch? *1 pt*

24 gal

k = 4 from the second color, then the whole, 6 parts; 12 found the first color and stopped

Scoring: 1 point.

19. 1 mi = 1.61 km. A train does 80 km/h; a bus does 45 mi/h. Which is faster? Show both speeds in one unit. *1 pt*

The train: 45 mi/h = 72.45 km/h, and 80 > 72.45.

Same unit first; 80 > 45 compared two units.

Scoring: 1 point.

Score Tracker

Benchmark	Questions	Possible	Earned
MA.7.GR.1.5	1, 2, 3, 4, 5, 6, 7, 8	10	
MA.7.AR.3.3	9, 10, 11, 12, 13, 14, 15	8	
MA.7.AR.4.5	16, 17, 18, 19	6	
Total		**24**	

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.

Multi-Step Proportional Relationships

24 points **WORKED ANSWERS**

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors *10 points*

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2. A 12 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.5

copy ÷ original; the copy is smaller, so k is under 1 — 2 is the proportion backwards

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All three and nothing else for the point: sides and perimeter × k, area × k^2, angles unchanged.

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a. Find the copy's perimeter. *1 pt*

32 cm

perimeter × k

b. Find the copy's area. *1 pt*

60 cm²

area × k² — 30 multiplied the area by k once

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